

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ **LATEST NTA PATTERN**
- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
- ✓ **DPP, REVISION SHEETS**
- ✓ **100% MONEYBACK TO MERITORIOUS STUDENTS**

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Designed by
Math Maestro

Anup Sir



JEE ADV. CRASH COURSE

TARGETING JEE ADVANCED 2020

- ✓ CONCEPT BUILDING VIDEOS
- ✓ ADVANCED LEVEL VIDEOS
- ✓ 2 LEVELS OF SHEETS,
1 WITH ALL TYPE OF QUES.
- ✓ LIVE DISCUSSION OF
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JEE Mains 2020 Jan Chapter wise Question Bank

Limits

Q1

Evaluate $\lim_{x \rightarrow 2} \frac{3^x + 3^{x-1} - 12}{3^{\frac{x}{2}} - 3^{1-x}}$

7th Jan Morning

Sol

72

Put $3^{\frac{x}{2}} = t$ रखने पर

$$\Rightarrow \lim_{t \rightarrow 3} \frac{\frac{4t^2}{3} - 12}{-\frac{3}{t^2} + \frac{1}{t}} = \lim_{t \rightarrow 3} \frac{4(t^2 - 9)t^2}{3(-3 + t)} = \lim_{t \rightarrow 3} \frac{4t^2(3 + t)}{3} = \frac{4 \times 9 \times 6}{3} = 72$$

02

$\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ is equal to

$\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ का मान है—

(1) e^{-2}

(2) e^2

(3) $e^{2/7}$

(4) $e^{3/7}$

8th Jan Morning

Sol

(1)

$$\text{Let } L = \lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}} = e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{3x^2 + 2}{7x^2 + 2} - 1 \right\}} = e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{-4x^2}{7x^2 + 2} \right\}} = e^{\frac{-4}{2}} = e^{-2}$$

03

Image of $(1, 2, 3)$ w.r.t a plane is $\left(\frac{-7}{3}, \frac{-4}{3}, \frac{-1}{3}\right)$ then which of the following points lie on the plane

$(1, 2, 3)$ का किसी समतल में प्रतिबिम्ब $\left(\frac{-7}{3}, \frac{-4}{3}, \frac{-1}{3}\right)$ तो निम्न में से कौनसा बिन्दु समतल पर स्थित नहीं है :

- (1) $(-1, 1, -1)$ (2) $(-1, -1, -1)$ (3) $(-1, -1, 1)$ (4) $(1, 1, -1)$

8th Jan Evening

Sol

(4)

d.r of normal to the plane

समतल के लम्बवत् द्विकअनुपात

$$\frac{10}{3}, \frac{10}{3}, \frac{10}{3}$$

$$1, 1, 1$$

midpoint of P and Q is $\left(\frac{-2}{3}, \frac{1}{3}, \frac{4}{3}\right)$

P तथा Q का मध्य बिन्दु $\left(\frac{-2}{3}, \frac{1}{3}, \frac{4}{3}\right)$

equation of plane $x + y + z = 1$ समतल का समीकरण $x + y + z = 1$

04

If $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$ then the value of x at which $f(x) = [x^2] \sin \pi x$ is discontinuous

(where $[.]$ denotes greatest integer function)

यदि $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$ तब x का मान जिसके लिये $f(x) = [x^2] \sin \pi x$ असतत् हो

(जहाँ $[.]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है)

- (1) $\sqrt{A+1}$ (2) $\sqrt{A+21}$ (3) $\sqrt{A}$ (4) $\sqrt{A+5}$

9th Jan Evening

Sol

Limits

JEE Mains 2020 Jan Chapter wise Question Bank

(1)

$$\lim_{x \rightarrow 0} x \left(\frac{4}{x} - \left\{ \frac{4}{x} \right\} \right) = A \Rightarrow \lim_{x \rightarrow 0} 4 - x \left\{ \frac{4}{x} \right\} = A \Rightarrow 4 - 0 = A$$

check when जाँचने पर

(A) $x = \sqrt{A+1} \Rightarrow x = \sqrt{5} \Rightarrow$ discontinuous असतत्

(B) $x = \sqrt{A+21} \Rightarrow x = 5 \Rightarrow$ continuous सतत्

(C) $x = \sqrt{A} \Rightarrow x = 2 \Rightarrow$ continuous सतत्

(D) $x = \sqrt{A+5} \Rightarrow x = 3 \Rightarrow$ continuous सतत्



mathongo

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